SS-S-111C
27 April 1984
SUPERSEDING
SS-S-111B
18 May 1981

FEDERAL SPECIFICATION

SOUND CONTROLLING MATERIALS (TROWEL AND SPRAY APPLICATIONS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers acoustical materials for trowel or spray applications.

1.2 Classification. The acoustical materials shall be of the following types, densities, grades, and classes as specified (see 6.2):

Type I - Cementitious materials

Density A - 18 to 20 pounds per cubic foot (ppcf)
Density B - Over 20 ppcf

Type II - Fibrous Materials

Density C - 11 to 14 ppcf
Density D - Over 14 ppcf

Grades 1 through 7 according to level of noise reduction required (see 3.4.1)
Grades 101 through 107 according to level of sound absorption required (see 3.4.2)

Class (a) - Noncombustible (see 3.5.1)
Class (b) - Fire resistant (se 3.5.2)
Class (c) - Slow burning (see 3.5.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

UU-S-48 - Sacks, Shipping, Paper

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Federal Standard:

FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal activities may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C423 - Sound Absorption of Acoustical Materials in Reverberation Rooms, Test for

E84 - Marking for Shipment and Storage

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall supply containers of each type, density, grade, and class of acoustical material for first article inspection and approval (see 4.2.1 and 6.4).

3.2 Standard commercial product. The acoustical material of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the acoustical material being furnished. A standard commercial product is a product which, has been sold or is currently being offered for sale, on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Materials. Materials used shall be free from defects which would adversely affect performance. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all material incorporated in the work covered by this specification are to be new using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3.1 Type I. Type I ingredients shall be uniformly mixed and ready for use with the addition of water. The material in the container shall be dry and shall not be caked or sticky. After the ingredient has been properly mixed with the amount of water specified by the manufacturer, the materials shall be of such a consistency that it can be applied with a plastering trowel or with a suitable plaster-spraying machine. Upon drying, the acoustical material shall have a hardness adequate to form a durable surface.

3.3.2 Type II. Type II shall consist of mineral or vegetable fiber with an integrally mixed binder, or it may be supplied for use with a liquid binder added at the time of application.

3.4 Acoustical properties.

3.4.1 Noise-reduction coefficient (NRC). Acoustical materials shall be classified in terms of sound-absorbing efficiencies at four frequencies: 250, 500, 1000, and 2000 Hertz (Hz). The average of the sound-absorption coefficients at these four frequencies will be called the NRC. For grading purposes, the average will be taken to the nearest multiple of 0.05, with the provision that exact midpoints be reported as the next higher multiple of 0.05. For example, 0.625 and 0.675 would be reported as 0.65 and 0.70, respectively. The value of the coefficient shall be determined in accordance with 4.4.1 and shall comply with the requirements of table I (see 6.2).

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Table I. Grades for noise-reduction coefficient

Grade	NRC
1	0.90 and over
2	0.80 and 0.85
3	0.70 and 0.75
4	0.60 and 0.65
5	0.50 and 0.55
6	0.40 and 0.45
7	Less than 0.40

3.4.2 Sound-absorption coefficient. For auditorium treatments, acoustical material specified in the invitation for bids shall be graded in terms of sound-absorbing qualities at 500 Hz. The limiting values of sound-absorption coefficients at 500 Hz for each grade shall be determined in accordance with 4.4.1 and shall comply with the requirements of table II (see 6.2).

Table II. Grades for sound-absorption coefficients at 500 Hz (for auditorium treatment).

Grade	Coefficient at 500 Hz
101	0.90 and over
102	0.80 - 0.89
103	0.70 - 0.79
104	0.60 - 0.69
105	0.50 - 0.59
106	0.40 - 0.49
107	Less than 0.40

3.5 Surface-flame spread. When tested in accordance with 4.4.2, acoustical material shall be designated as follows (see 6.2).

3.5.1 Class (a), noncombustible. The flame-spread index of the acoustical material shall not exceed 25 nor shall the material show evidence of continued progressive combustion. Materials with a flame-spread index of 25 or less are considered noncombustible (see 6.2).

3.5.2 Class (b), fire resistant. The flame-spread index shall not exceed 75 on the exposed surface. Materials with a flame-spread index of 75 or less are considered fire resistant (see 6.2).

3.5.3 Class (c), slow burning. The flame-spread index shall not exceed 200 on the exposed surface. Material having a flame-spread index of 200 or less is considered slow burning. Materials having a flame-spread index of over 200 are considered combustible (see 6.2).

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3.6 Dust test. The acoustical material shall be subjected to the dust test specified in 4.4.3. Dust removed from the acoustical material shall not exceed 0.025 gram per square foot. Unless otherwise specified (see 6.2), the dust test shall not be required for Civil Agency procurement.

3.7 Certificate of compliance. Unless otherwise specified (see 6.2), acoustical material offered under this specification shall be certified by the contractor to have successfully completed the tests for sound absorption and noise-reduction, flame-spread index, and dust.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. First article inspection shall be performed on one acoustical material sample when a first article sample is required (see 3.1, 6.2, and 6.4). This inspection shall include the tests of 4.4. Failure of the first article to pass the tests shall be cause for rejection. The first article may be a standard production item from the contractor's current inventory provided the material meets the requirements of the specification and is representative of the manufacturing technique applicable to the remaining material to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on samples selected from each lot offered to the Government. The inspection shall include the tests of 4.4, and the packaging inspection of 4.5.

4.3 Inspection lot. All acoustical material of the same type, density, class, and grade offered for delivery at one time shall be considered a lot for purposes of inspection.

4.4 Test methods.

4.4.1 Sound absorption and NRC test. Each sample for sound absorption and noise reduction tests shall consist of enough acoustical material, ready for application when mixed with water or binder as applicable to cover an area of approximately 62 square feet (8 feet by 9 feet). Sound absorption and NRC of the acoustical sample material shall be determined in accordance with ASTM C423.

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4.4.2 Flame-spread test. The sample for flame-spread test shall consist of enough material, ready for application with the possible exception of water only, to cover an area approximately 50 square feet. The flame-spread index shall be determined in accordance with ASTM E84.

4.4.3 Dust test. The sample for the dust test shall consist of enough material, ready for application with the exception of water, to cover an area of not less than 4 square feet.

4.4.3.1 Specimen. The sample of acoustical material in the type and density specified shall be applied to a flexible flat 3/8 inch thick asbestos-cement sheet. Application shall be in strict accordance with the manufacturer's printed instructions, including any prescribed surface preparation and finish treatment, except that not less than 0.5 inch of acoustical material shall be applied. Specimen shall be conditioned at 73 degree (°) ± Fahrenheit (F) and 50 ± 5 percent relative humidity until the weight of the specimen, weighed at 24-hour intervals, becomes constant to within 0.1 percent.

4.4.3.2 Test apparatus. The test apparatus shall consist of a tight metal duct approximately 17 inches wide by 24 inches high by 7 feet long. A centrifugal blower, driven by a 7.5 horsepower motor, shall be coupled to the duct. One side of the duct shall have an opening large enough to insert a test sample 16 inches wide by 48 inches long, having approximately 4 square feet of surface area exposed to the air flow (see figure 1). A vane anemometer shall be mounted in front of the test panel area for measurement of the air flow velocity. Two dust-collecting filters, having a 95 percent efficiency in filtering particles 5 microns in size and larger, shall be installed upstream of the opening for the test sample. A black dust-collecting cloth shall be placed downstream of the opening for the test sample. The cloth fabric shall be made of 30 denier nylon having a thread count of 102 warp ends per inch and 82.3 filler picks per inch. The cloth shall be dried at 120° ± 1°F to a constant weight before each weighing.

4.4.3.3 Procedure. The test for dusting shall be conducted in a reasonably dust-free environment. The test specimen shall be inserted in the test opening in the duct side wall. The complete system shall be purged for 6 hours at a minimum impinging velocity of 800 feet per minute over the entire surface of the test specimen. After the system has been purged for 6 hours, the dust cloth shall be removed and dried at 120°F and weighed until the weight is constant with the nearest 0.001 gram. The dust-collecting cloth shall be reinserted in the test apparatus and the system again purged for 24 hours. The dust-collecting cloth shall then be removed, dried, and weighed as before. The net difference in weight between the 6-hour purging and the 24-hour purging shall constitute the dust removed from the acoustical material. The weight of the dust removed shall not exceed the amount specified in 3.6.

4.5 Preparation for delivery inspection. An inspection shall be made to determine that the packing and marking comply with the requirements in section 5 of this specification. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping sacks or containers in the end item inspection lot. The inspection level shall be S-2 with an Acceptable Quality Level of 4.0 percent defective.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Levels A and B.

5.1.1.1 Types I and II. Material of one type shall be furnished in sacks conforming to the applicable requirements of UU-S-48 for building plaster as indicated for sack construction number applicable to the weight involved.

5.1.2 Commercial. The material shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with applicable carrier rules and regulations.

5.2 Marking.

5.2.1 Civil agencies. Marking shall be in accordance with FED-STD-123.

5.2.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Acoustical materials are intended for use in the control of reverberation in auditoriums or other rooms. Type I material is normally applied over gypsum and portland cement base coats, and over metal laths and clip-systems. Type II materials are intended for spraying beams, roof decks, and columns.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type, density, grade, and class of acoustical material required (see 1.2, 3.4 through 3.5)
c. When a first article is required; type, density, grade, and class required (see 3.1, 4.2.1, and 6.4)
d. If dust test is required by Civil Agencies (see 3.6)
e. Certification of test data if different (see 3.7)
f. Level of packing required (see 5.1)

6.3 Contract Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104-9(n) of the Defense Acquisition Regulations (DAR) the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104-9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104-55 of the DAR. The first article should be first a production item, or it may be a standard production item from the contractor's current inventory as specified

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in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangement for examinations, tests, and approval of the first article.

6.5 Applications of materials. To insure proper and uniform thickness, it is recommended that acoustical materials be applied between spot ground of proper size, space at intervals of about 8 feet, or by other means recommended by the manufacturer to give equivalent results. After application, the acoustical material should be finished to a true surface with a cork float, steel trowel, roller, or other means.

6.6 Paintability. When acoustical material covered by this specification is to be painted, a nonbridging paint should be used to prevent damage to the sound-absorption property.

Custodians:

Army - ME
Navy - YD

User Activity:

Army - CE

CIVIL AGENCY COORDINATING ACTIVITIES

GSA-FSS-PCD
VA-OSS

PREPARING ACTIVITY

Navy - YD

DoD Project 5610-0199

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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